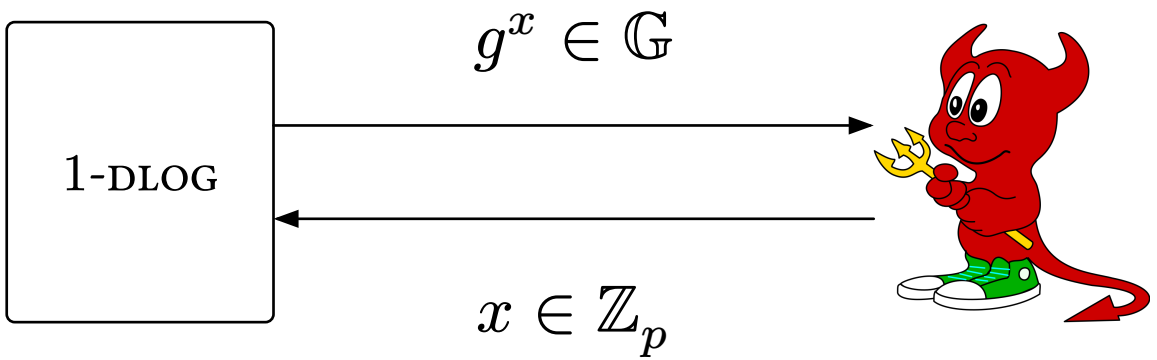
Generic Bounds for Multi-Instance Problems: The Strange Case of Inverse Diffie–Hellman

Akshima, Eike Kiltz, Aysan Nishaburi,
Samin Nooripoor, **Emiel Wiedijk**

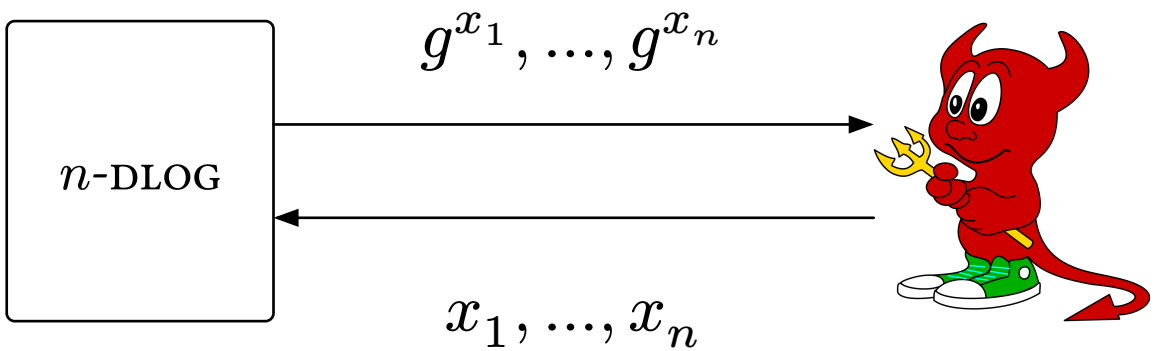
Ruhr University Bochum

Discrete logarithm



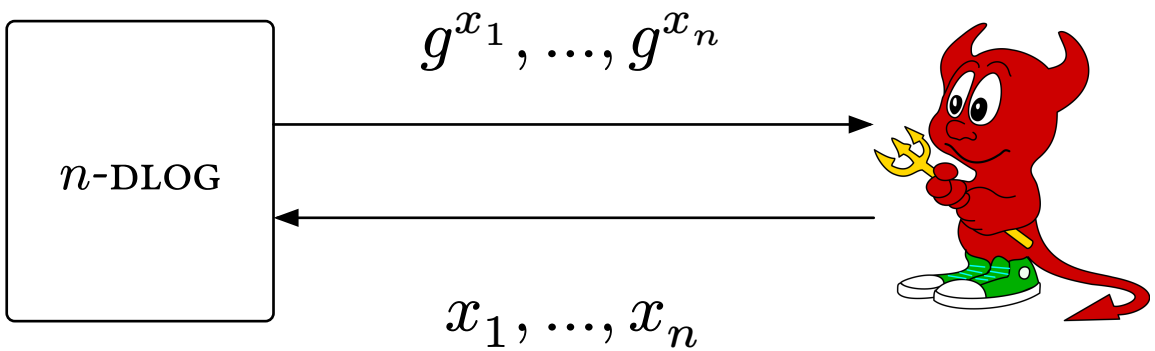
where $\mathbb{G}$ cyclic group of prime order p with generator g.

Multi-instance assumptions



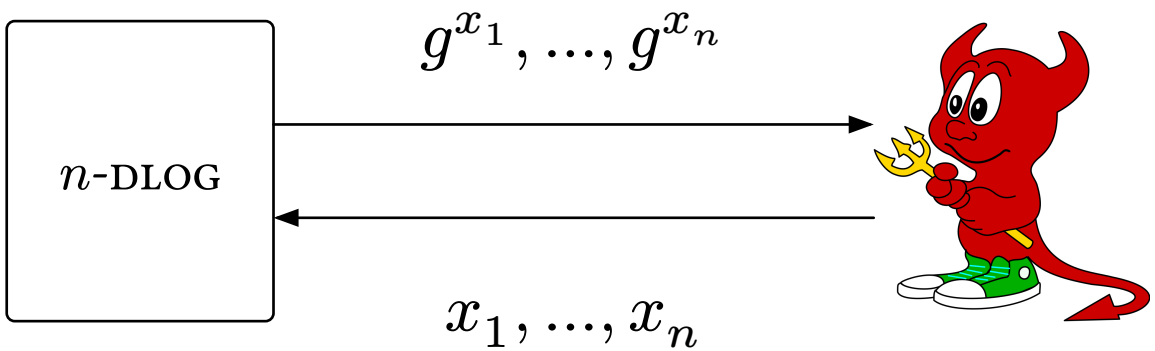
Applications: multi-user settings, better concrete security for PAKE [Bao+26], time-space tradeoffs.

Multi-instance assumptions



Question: How much harder is n-dlog, compared to 1-dlog?

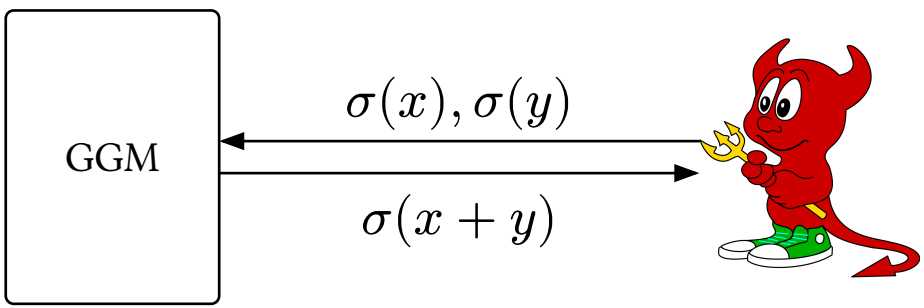
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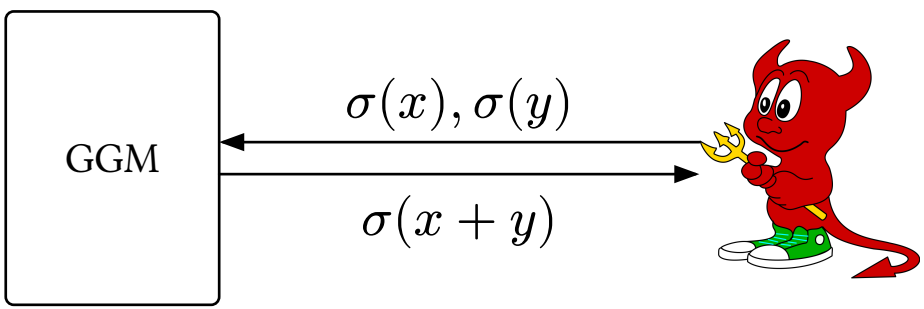
Hope: Solving n instances takes n times as long.

Generic Group Model/GGM [Sho97]



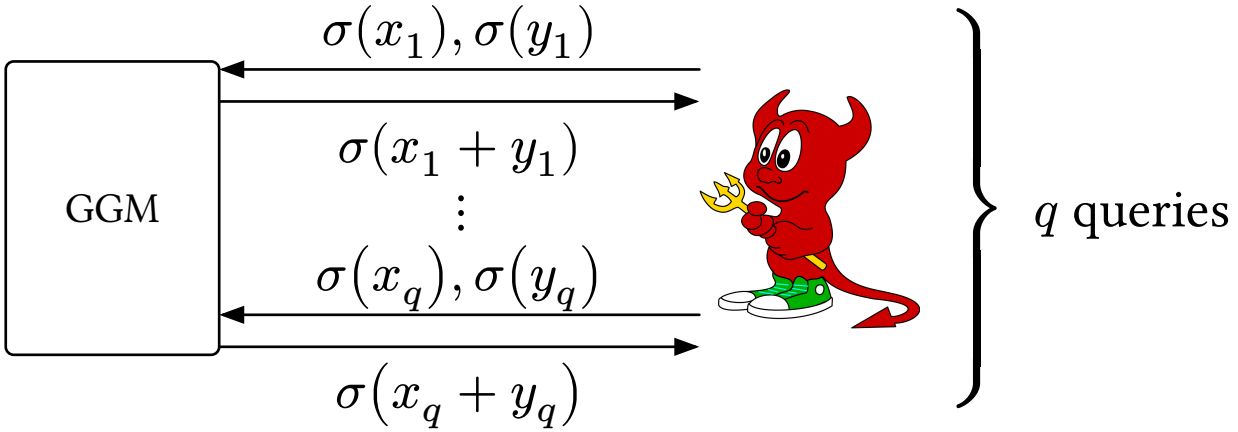
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- **GGM**: g^x represented by $\sigma(x)$, for random injection $\sigma : \mathbb{Z}_p \to \mathcal{L}$.

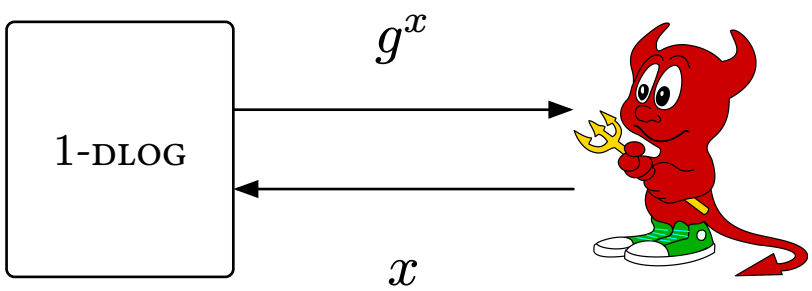
Generic Group Model/GGM [Sho97]



- Unconditional lower bounds are hard: use an **idealized model**.
- **GGM**: g^x represented by $\sigma(x)$, for random injection $\sigma : \mathbb{Z}_p \to \mathcal{L}$.
- **Compute** $\mathsf{Adv}(q)$: optimal advantage given q queries

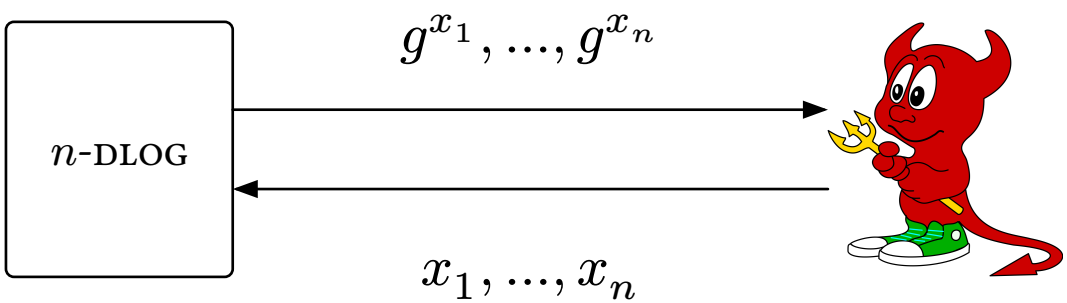
Hardness of discrete logarithm in GGM

1 instance



$q \approx \sqrt{p}$ group operations [Sho97]

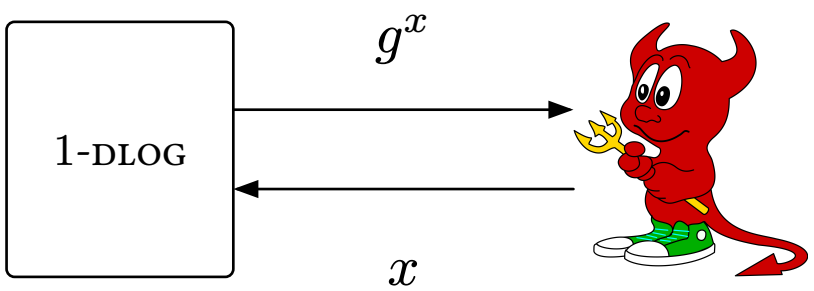
n instances



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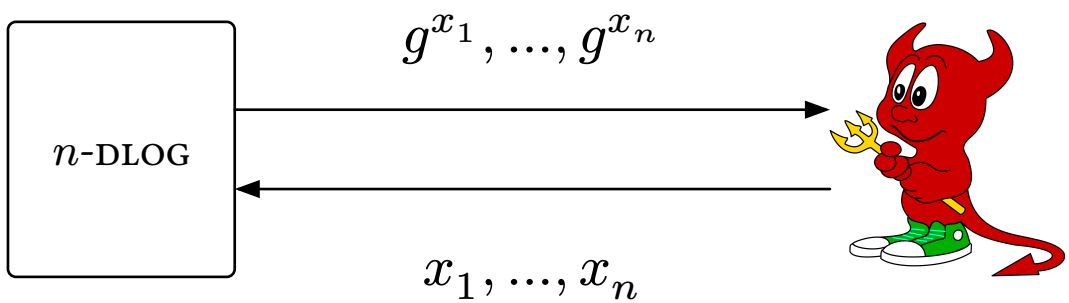
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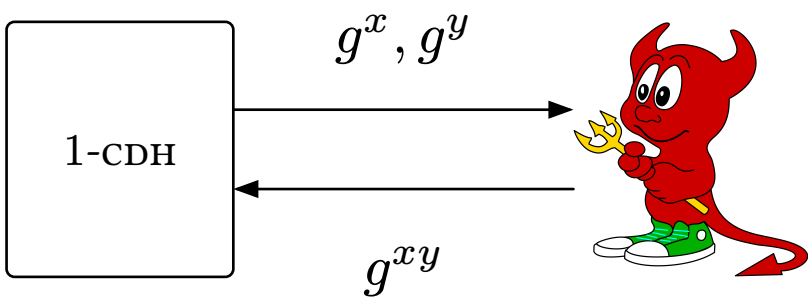


$q \approx \sqrt{np}$ group operations [Yun15]

So: n-dlog takes $\sqrt{n}$ times longer. For which other problems?

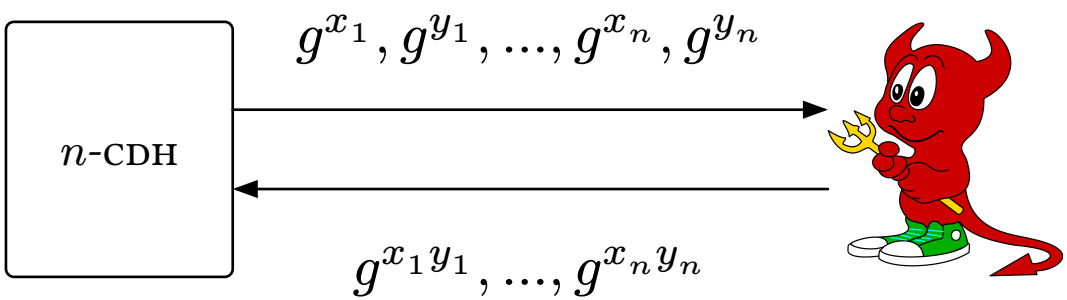
Other problems: (computational) Diffie–Hellman

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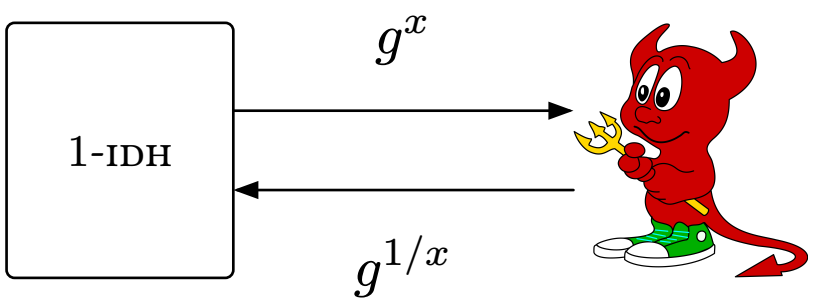
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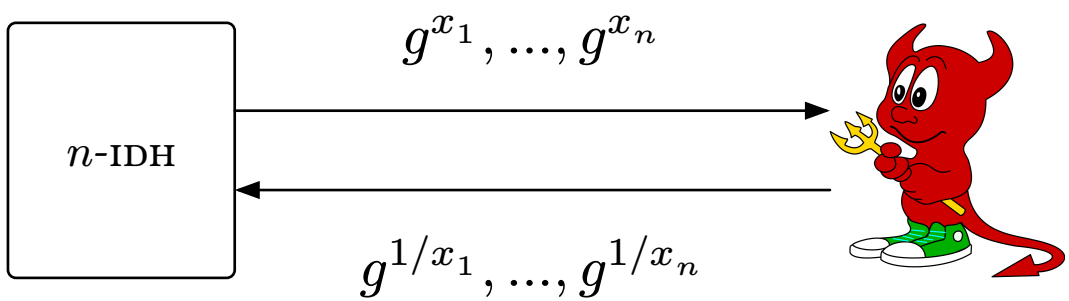
Other problems: Inverse Diffie–Hellman (IDH)

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$q \approx \sqrt{p}$ group operations

n instances



$q \approx$ **?** group operations

IDH is **equivalent to CDH** in single-instance [SS01].

Our contributions

1. The n-MI_W abstraction, capturing multi-instance CDH/IDH,
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1. The n-MI_W abstraction, capturing multi-instance CDH/IDH,
   - *Also k-out-of-n and with discrete-log (corruption) queries*
2. GGM bounds of n-MI_W based on **guessing advantage**
3. Find concrete GGM bounds for
   - Multi-instance Computational/Square Diffie–Hellman
   - Multi-instance Inverse Diffie–Hellman
   - *Some variants of Kernel Diffie–Hellman [MRV16]*

GGM bounds for n-CDH/n-IDH

n-CDH

$$\mathsf{Adv}_{n\text{-CDH}}(q) \approx \left(\frac{q^2}{np}\right)^n$$

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Corollary. At least $\approx \sqrt{np}$ queries are needed for constant advantage.

Guessing advantage

- **Guessing advantage** (informal): probability that adversary gives correct output **without looking at the input**

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- Example: CDH given g^x, g^y compute g^{xy}.

$$\text{guessing advantage} := \max_{\alpha,\beta,\gamma} \Pr_{x,y \leftarrow \mathbb{Z}_p} [\alpha x + \beta y + \gamma = xy].$$

- For single-instance search problems: mostly trivially $\approx \frac{1}{p}$.

Guessing advantage for multi-instance problems

- n-instance problems: intuitively guessing advantage $\frac{1}{p^n}$
- For n-CDH: guess each $x_i y_i$ as a linear combination of all inputs,

  $$\text{guessing advantage} := \max_{A,B \in \mathbb{Z}_p^{n \times n}, c \in \mathbb{Z}_p^n} \Pr_{\boldsymbol{x},\boldsymbol{y} \leftarrow \mathbb{Z}_p^n} [A\boldsymbol{x} + B\boldsymbol{y} + c = \boldsymbol{x} \odot \boldsymbol{y}],$$

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- After **algebraic geometric** analysis: guessing advantage $\approx \frac{1}{p^n}$

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- **Algebraic geometry** shows this is tight.

From guessing advantage to GGM bounds

Theorem (informal).

$$\mathsf{Adv}(q) \lesssim \sum_{i=0}^{n} \binom{q^2}{i} \left(\frac{2}{p}\right)^i \cdot \mathrm{Guess}(n-i),$$

where $\mathrm{Guess}(m)$ is the guessing advantage for m instances.

From guessing advantage to GGM bounds

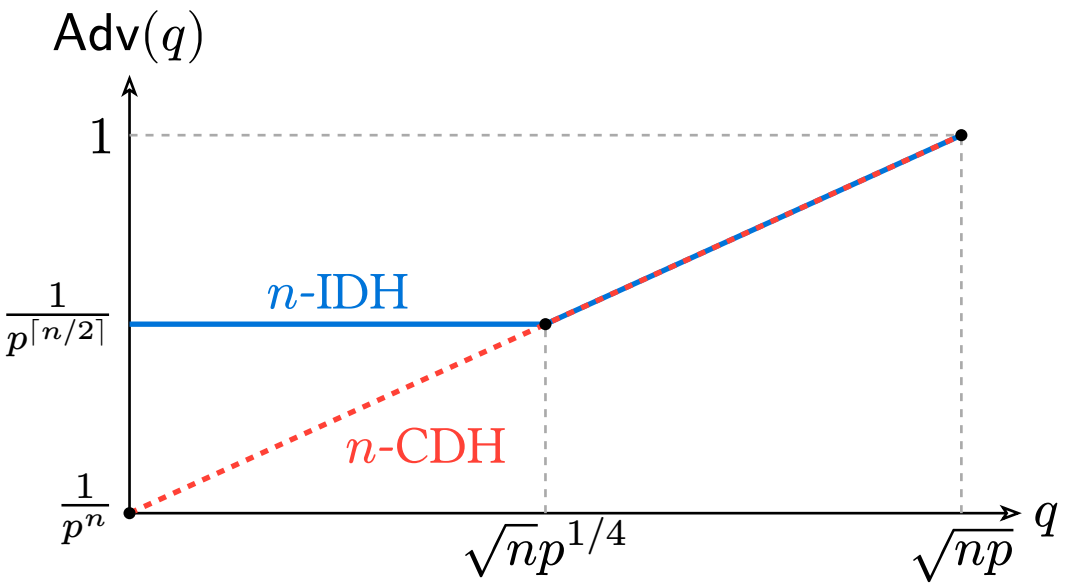
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	$\mathrm{Guess}(m)$	$\mathsf{Adv}(q)$
n-CDH	$1/p^m$	$\left(q^2/(np)\right)^n$
n-IDH	$1/p^{\lceil m/2 \rceil}$	$\max\left(\left(q^2/(np)\right)^n, 1/p^{\lceil n/2 \rceil}\right)$

n-CDH vs. n-IDH


Adv(q)
1
$\frac{1}{p^{\lceil n/2 \rceil}}$
$\frac{1}{p^n}$
n-IDH
n-CDH
$\sqrt{n}p^{1/4}$
$\sqrt{np}$
q

Conclusion

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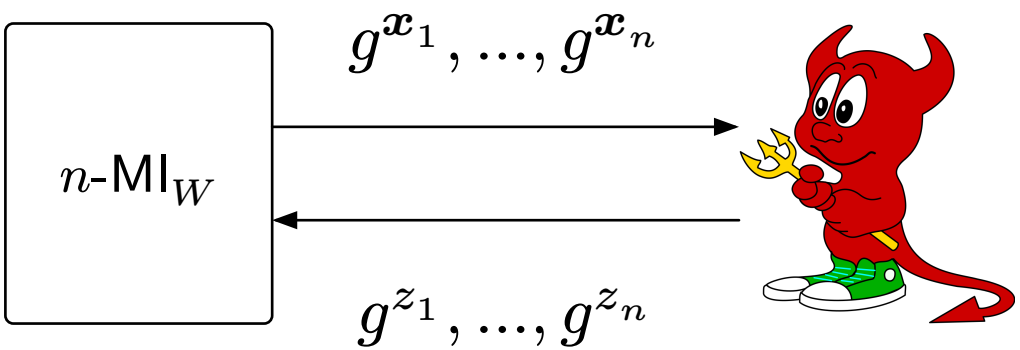
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Thank you for your attention!

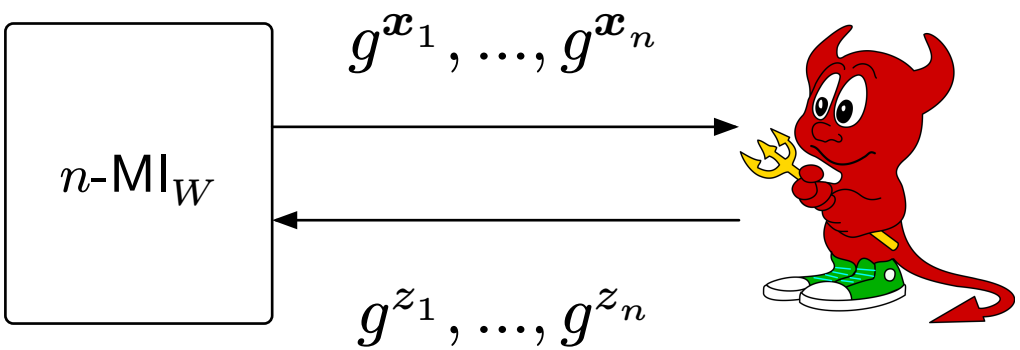
The n-MI_W abstraction



Problem	x_i	$W(x_i, z_i)$
CDH	(x_i, y_i)	$x_i y_i - z_i$
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- Adversary wins iff $W(\boldsymbol{x}_i, z_i) = 0$ for **all** $i \in [n]$.

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